Case Study 2 Report

```
parsed http://www.basketball-reference.com/leagues/NBA_2025_advanced.html
Advanced
```

0.1 STA540 Case Study 2 Draft

Project Title: All-NBA Selection Prediction

Data citation: Sports Reference LLC "2024-25 NBA Player Stats: Totals" Basketball-Reference.com - Basketball Statistics and History. https://www.basketball-reference.com/

Original data collection date:

Data pull date: 02/03/2026

Acronyms:

- NBA - National Basketball Association
- GLMM - generalized mixed effect model
- AUC - Area under the curve
- PR - precision-recall
- ROC - receiver operating characteristic
- WS - win share

0.1.1 1 Introduction

Background/Introduction:

The All-NBA Team is one of the National Basketball Association's most prestigious individual honors. It is awarded to the players that have performed at the highest level during the regular season. The selections have been determined through voting by a panel of 100 sportswriters and broadcasters, with players ranked based on ballot points and assigned to first (best), second, and third teams. These awards not only count towards a player's legacy, but also their salary (contract incentives), reputation, and Hall of Fame consideration.

Advances in computing have led to new basketball metrics and analysis. Advanced metrics like win share attempt to quantify player contribution in one number to objectively measure player performance. Modelling all-NBA selections using objective metrics not only proves or disproves the value of some of these metrics, but also showcases the impact subjective measures like voter bias or popularity have on NBA awards.

1.1 Study Aims

The study will examine player box scores throughout a season and calculate advanced statistics, namely win share, to identify if All-NBA status can be predicted from metrics using a generalized linear mixed effect model (GLMM). The primary aim of the study will be to determine the predictive value of win shares on all-NBA selection. Secondary aims will look at other advanced metrics like box-plus-minus and compare those with win share.

0.1.2 2 Data

The data for this study comes from Basketball Reference, a website that aggregates several basketball statistics. The key variables of interest are `is_all_nba`, a binary variable indicating whether a player was selected, and `ratioWS`, which is a number representing the win share of the player. Win share is a metric that attempts to divide up a team's wins to its player proportionally by contribution (https://www.basketball-reference.com/about/ws.html).

The dataset includes NBA players with 65 games played and average over 20 minutes in those games, based on selection criteria for the award. There is no missing data, due to meticulous stat-tracking.

0.1.3 3 Exploratory Data Analysis

In Figure 1, there is a clear distinction in win shares across the 4 different selection groups. This indicates that win shares can be used as a useful predictor. There is, however, some overlap between the all-NBA groups and some of the unselected players, meaning that win shares alone cannot predict all-NBA status.

0.1.4 4 Model Fitting and Diagnostics

4.1 Statistical Model

$logit(p_{ij}) = X_{ij}\beta + u_j$

where:

- $logit(p_{ij}) = log(p_{ij}/(1 - p_{ij}))$ corresponding to the log odds ratio
- X_{ij} is the fixed effect features (win share, minutes played, etc…)
- β is the corresponding coefficients
- u_j is the random effect corresponding to each team

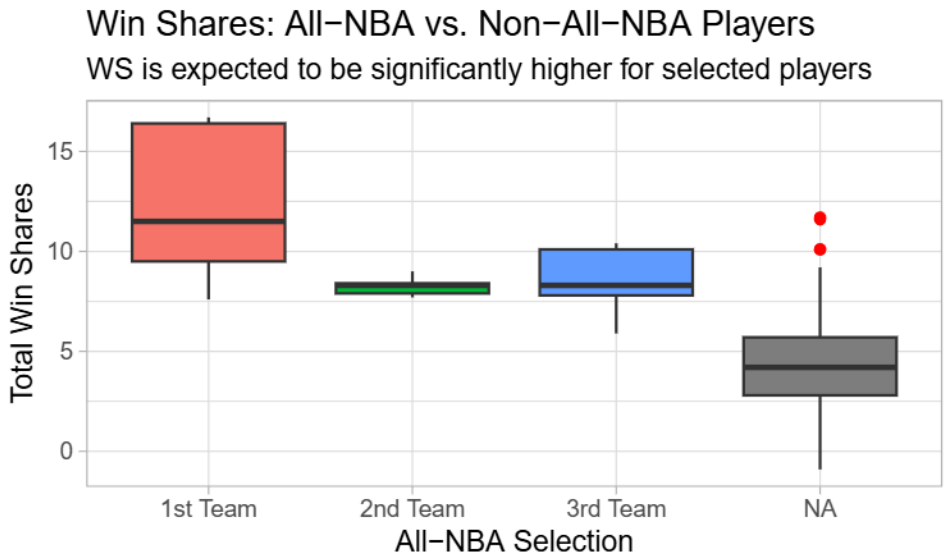


Figure 1

4.2 Rationale

Logistic regression is commonly used for binary response variables. This GLMM extends that idea by accounting for random effects of clustering as well. For instance, there may be unobserved heterogeneity between teams that are not captured in the player data itself. Some examples are market size, voter biases, or a winning record. This model accommodates the hierarchical nature of the data that is clustered by team. Instead of the typical threshold, the top 15 highest probabilities will be chosen.

0.1.5 5 Discussion

The model assumes independent observations conditional on the predictors, but if one player makes the All-NBA team, it lowers the chances of another making the team. In addition, voters can be influenced by factors outside of in-game box stats, such as: news (narratives), popularity, and strength of schedule. This would cause omitted variable bias.

Because the initial frequentist model did not converge, I fit a Bayesian logistic regression with a horseshoe prior to stabilize estimation under multicollinearity and to shrink weak predictors toward zero. All predictors were standardized, so each coefficient corresponds to a one standard deviation increase in the predictor. Posterior sampling mixed well (all $\hat{R} \approx 1.00$ and effective sample sizes > 1000), indicating convergence and stable posterior estimates. The posterior suggests that **points scored (PTS)** and **Win Share ratio (ratioWS)** are the strongest predictors of All-NBA selection. **PTS** has a large positive association with selection probability ($\beta = 4.15$, 95% CrI: 1.82 to 7.97), indicating that higher scoring substantially increases the log-odds of being selected, even after controlling for other box score variables. **ratioWS** is also strongly positive ($\beta = 2.86$, 95% CrI: 0.85 to 5.94), supporting the primary aim that Win Shares contain meaningful predictive signal for All-NBA outcomes. Other variables such as minutes played (MP), steals (STL), blocks (BLK), and personal fouls (PF) had posterior intervals overlapping zero in this fit,

suggesting weaker or uncertain incremental predictive value once scoring and Win Shares are included.

This case study demonstrates that objective metrics can predict All-NBA selection with meaningful accuracy, with **PTS** and **ratioWS** emerging as the most informative predictors among those considered. However, the modeling goal is fundamentally constrained by the award structure (only 15 selections) and by unmeasured factors such as media narratives, player popularity, and contextual team performance. Future work could incorporate team record, betting odds/award narratives, and position/role indicators, and could evaluate performance using cross-validation and precision-recall metrics tailored to class imbalance.

0.1.6 6 References

1. Popovic, G. (2022). *Generalised mixed models.* Environmental Computing. https://environmentalcomputing.net/statistics/mixed-models/mixed-model-3/

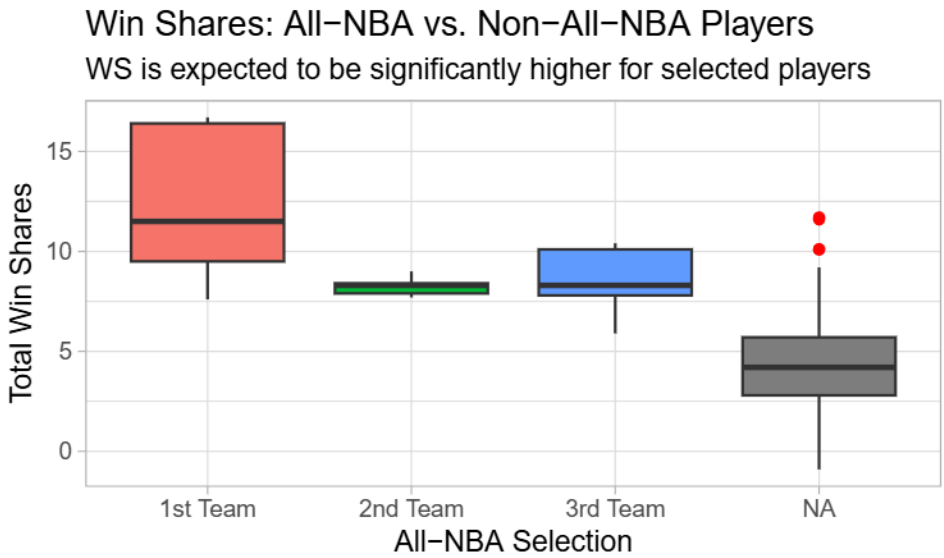
2. Bresler A (2026). _nbastatR: R's interface to NBA data_. R package version 0.1.153, commit f1e25e746f8a9dffdd7cc4eb3666efb37fc6a8b9, <https://github.com/abresler/nbastatR>.

0.2 Appendix

variable	mean	sd	median	p90
Rk	109.7733333	74.4002550	101.0000	220.7000
Age	26.3266667	4.4306690	26.0000	32.0000
G	73.7600000	4.8710623	74.0000	80.0000
GS	48.9266667	26.7187933	56.5000	77.0000
MP	2049.9933333	421.1789382	2068.5000	2610.4000
FG	369.6800000	156.1802143	338.5000	592.0000
FGA	780.1133333	313.7681685	691.0000	1226.5000
FG%	0.4747467	0.0614084	0.4610	0.5572
3P	122.0533333	69.0787690	122.0000	216.0000
3PA	330.9333333	175.3304710	330.5000	553.9000
3P%	0.3484762	0.0770183	0.3650	0.4160
2P	247.6266667	143.5212748	209.5000	441.5000
2PA	449.1800000	249.8170893	395.0000	801.2000
2P%	0.5473533	0.0577959	0.5430	0.6270
eFG%	0.5550200	0.0462946	0.5510	0.6154
FT	150.4400000	109.7475125	117.5000	296.2000
FTA	190.6266667	133.0363803	150.5000	366.4000
FT%	0.7834467	0.0867072	0.7955	0.8901
ORB	86.7000000	62.9029027	66.0000	185.5000
DRB	277.7066667	134.5284355	248.0000	443.4000
TRB	364.4066667	185.5802110	308.5000	623.4000
AST	246.3533333	155.2550678	178.5000	434.3000
STL	68.3533333	26.7519394	66.0000	103.0000
BLK	38.3666667	30.2087628	30.0000	85.4000
TOV	118.3733333	62.0845436	103.0000	202.0000
PF	149.8800000	39.6504729	150.5000	201.0000
PTS	1011.8533333	439.5624943	897.0000	1710.0000
Trp-Dbl	0.7933333	3.3404108	0.0000	1.0000
ratioWS	4.8793333	2.9539939	4.4000	8.4100
ratioOWS	2.7460000	2.3196456	2.4000	5.7100
ratioDWS	2.1326667	1.0172008	2.0000	3.8000
ratioWSPer48	0.1112467	0.0568971	0.1080	0.1749
ratioOBPM	0.3533333	2.4883802	0.0000	3.5000
ratioDBPM	0.0693333	1.2470070	0.0000	1.7100
ratioBPM	0.4106667	2.9561027	0.2000	4.0100
ratioVORP	1.3753333	1.7747270	1.2000	3.6000

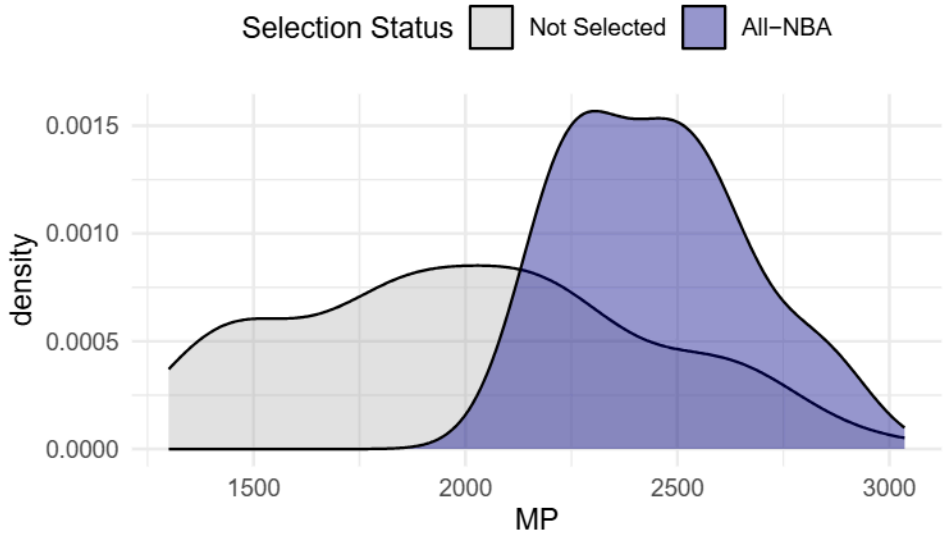
0.2.1 Win Share Compared to All-NBA Selection Status

Looking at the graph below, there is a clear distinction in win shares across the 4 different selection groups. This indicates that win shares can be used as a useful predictor. There is, however, some overlap between the all-NBA groups and some of the unselected players, meaning that win shares alone cannot predict all-NBA status.



0.2.2 Usage and All-NBA

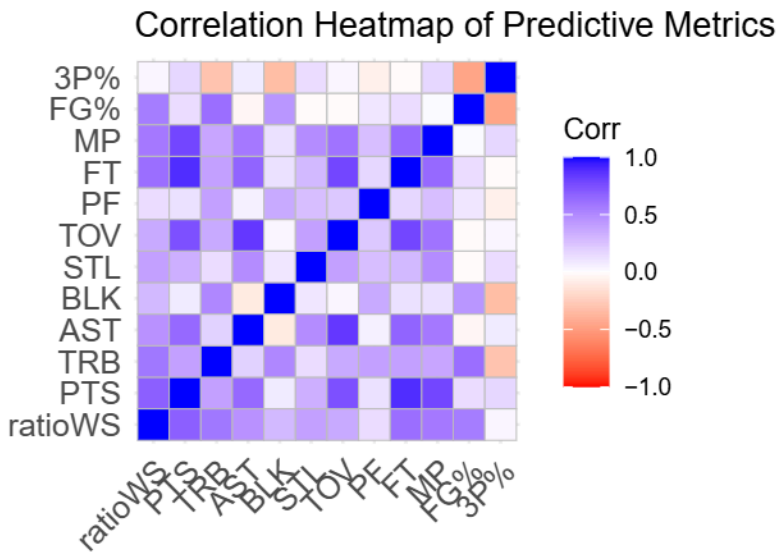
A common expression in the NBA is "availability is the best ability". The graph below shows that all-NBA players have higher minutes played compared to other players, indicating higher usage. The best players typically play the most minutes in a game, but there is not enough separation in the distribution to use minutes alone.



0.2.3 Correlation Plot With Selected Predictors

The correlation plot below shows that there is a moderate correlation between some of the key predictors. The largest of which is free throw attempts and points, which make sense

as they go hand in hand. There also also moderate correlations between minutes played and points, turnovers and assists, free throws and turnovers.



0.2.4 Logistic GLMM (Model 1)

0.2.4.1 Assumptions

The model assumptions are as follows (Popovic):

1. Observations are independent conditional on predictors and random effects.
2. The response variable follows a distribution from the exponential family (e.g., binomial, Poisson, Gaussian).
3. A link function (logit) defines a linear relationship between predictors and the mean response.
4. Random effects are independent of the response.
5. Random effects are typically normally distributed.

0.2.4.2 Model Assessment

To test the assumptions, the random effects should be approximately normally distributed. This can be checked with a Q-Q plot. The residuals should also not be correlated with the fixed effects, which can be checked with a scatter plot of the partial residuals. The Intraclass Correlation Coefficient (ICC) will provide a measure of what percentage of the variation is explained by the team rather than the individual stats.

The performance of the model can be evaluated using a AUC-PR metric, which measures the area under the the precision recall curve of the model. This metric works better for unbalanced data sets like this one with 10% of the data coming in the minority class. Other metrics that could also be useful are ROC AUC, f1 score, and accuracy.

Unfortunately, the initially proposed model did not converge. This means that the coefficients and the p-values cannot be interpreted.

```
Call:
glm(formula = is_all_nba ~ ratioWS + G + MP + FG + FGA + `3P` +
    FT + FTA + TRB + AST + STL + BLK + TOV + PF + PTS, family = binomial(link = "logit"),
    data = player_df)

Coefficients: (1 not defined because of singularities)
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  2.782e+02  1.230e+06   0.000    1.000
ratioWS      2.596e+01  5.630e+04   0.000    1.000
G           -6.050e+00  7.834e+03  -0.001    0.999
MP          -7.046e-02  2.456e+02   0.000    1.000
FG          -4.507e-01  1.818e+03   0.000    1.000
FGA          3.381e-01  9.000e+02   0.000    1.000
`3P`        -1.175e-01  1.008e+03   0.000    1.000
FT           2.888e-01  2.385e+03   0.000    1.000
FTA         -2.562e-01  1.729e+03   0.000    1.000
TRB          2.587e-03  3.199e+02   0.000    1.000
AST         -5.520e-02  3.399e+02   0.000    1.000
STL         -9.339e-02  4.439e+03   0.000    1.000
BLK          5.540e-01  8.250e+02   0.001    0.999
TOV          2.755e-01  5.855e+02   0.000    1.000
PF          -2.570e-01  9.121e+02   0.000    1.000
PTS                 NA         NA      NA       NA

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 9.7525e+01  on 149  degrees of freedom
Residual deviance: 1.4954e-08  on 135  degrees of freedom
AIC: 30

Number of Fisher Scoring iterations: 25
```

0.2.5 Logistic Regression with LASSO

Removing the random effect should help the algorithm converge, as it removes additional variables that don't have to be estimated. LASSO should help with variable selection as there is definitely multicollinearity in the equation as specified.

The coefficients provide a sense of magnitude and which variables are more useful in predicting the overall response variable.

```
16 x 1 sparse Matrix of class "dgCMatrix"
                      s0
```

```
(Intercept) 10.750976142
ratioWS      1.149600761
G           -0.463627956
MP           .
FG           .
FGA          0.007076230
`3P`         0.007808476
FT           .
FTA          0.003573613
TRB          .
AST          .
STL          0.016005497
BLK          0.030934729
TOV          0.007375041
PF          -0.009585528
PTS          .
```

0.2.6 Bayesian Logistic Regression with Horseshoe Prior

Let Y_i denote whether player i was selected to an All-NBA team.

$$Y_i \sim \text{Bernoulli}(p_i)$$

The probability of selection is modeled using a logistic link:

$$\text{logit}(p_i) = \beta_0 + \beta_1 \text{ratioWS}_i + \beta_2 G_i + \beta_3 MP_i + \beta_4 FG_i + \beta_5 FGA_i + \beta_6 FT_i + \beta_7 FTA_i + \beta_8 TRB_i + \beta_9 AST_i + \beta_{10} STL_i +$$

where all predictor variables are standardized.

To perform shrinkage and automatic variable selection, the regression coefficients are assigned a horseshoe prior:

$$\beta_j \sim \mathcal{N}(0, \tau^2 \lambda_j^2), \quad j = 1, \dots, 14$$

$$\lambda_j \sim C^+(0, 1)$$

$$\tau \sim C^+(0, 1)$$

where $C^+(0, 1)$ denotes the half-Cauchy distribution. The global parameter τ controls overall shrinkage, while the local parameters λ_j allow individual coefficients to escape shrinkage when strongly supported by the data.Let Y_i denote whether player i was selected to an All-NBA team.

$$Y_i \sim \text{Bernoulli}(p_i)$$

```
Running "C:/PROGRA~1/R/R-44~1.1/bin/x64/Rcmd.exe" SHLIB foo.c
using C compiler: 'gcc.exe (GCC) 13.3.0'
gcc  -I"C:/PROGRA~1/R/R-44~1.1/include" -DNDEBUG   -I"C:/Users/luxma/AppData/Local/R/win-li
cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
In file included from C:/Users/luxma/AppData/Local/R/win-library/4.4/RcppEigen/include/Eige
                 from C:/Users/luxma/AppData/Local/R/win-library/4.4/RcppEigen/include/Eige
                 from C:/Users/luxma/AppData/Local/R/win-library/4.4/StanHeaders/include/st
                 from <command-line>:
C:/Users/luxma/AppData/Local/R/win-library/4.4/RcppEigen/include/Eigen/src/Core/util/Macros
  679 | #include <cmath>
      |          ^~~~~~~
compilation terminated.
make: *** [C:/PROGRA~1/R/R-44~1.1/etc/x64/Makeconf:289: foo.o] Error 1


 Family: bernoulli
  Links: mu = logit
Formula: is_all_nba ~ ratioWS + G + MP + FG + FGA + FT + FTA + TRB + AST + STL + BLK + TOV
   Data: player_df_scaled (Number of observations: 150)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept    -9.49      3.54   -18.09    -4.76 1.00     1966     1111
ratioWS       3.29      1.64     0.77     7.07 1.00     2289     2253
G            -1.97      1.10    -4.49    -0.16 1.00     2019     1781
MP           -1.12      1.35    -4.28     0.79 1.00     3641     3457
FG            0.61      1.39    -1.88     3.73 1.00     3499     3020
FGA           1.23      1.57    -1.09     4.90 1.00     2859     2541
FT            0.48      1.18    -1.68     3.12 1.00     3709     2540
FTA           0.39      1.15    -1.68     2.95 1.00     3896     3051
TRB          -0.06      0.63    -1.43     1.23 1.00     3845     3504
AST           0.20      0.70    -1.18     1.74 1.00     3399     3174
STL           0.54      0.72    -0.70     2.18 1.00     3223     3439
BLK           0.79      0.76    -0.32     2.52 1.00     2575     2238
TOV           0.35      0.81    -1.07     2.25 1.00     3512     3283
PF           -0.65      0.64    -2.08     0.32 1.00     3127     3455
PTS           1.47      1.74    -1.08     5.61 1.00     2535     2547

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).


Probability of Direction

Parameter   |     pd
--------------------
```

```
(Intercept) |   100%
ratioWS     | 99.75%
G           | 99.05%
MP          | 81.15%
FG          | 68.67%
FGA         | 80.00%
FT          | 66.03%
FTA         | 64.03%
TRB         | 54.45%
AST         | 62.68%
STL         | 78.10%
BLK         | 86.92%
TOV         | 67.12%
PF          | 86.50%
PTS         | 81.67%
```

0.2.6.1 Assumptions and Assessment

The model diagnostics look quite good as the Rhat for each parameter is 1 or extremely close to 1, indicating convergence. Both the bulk and tail effective sample size is greater than 1000, which means that the sequence explored the sample space well.

The model also manages to correctly identify all 15 all-NBA players. The ranks for each player also closely match the 1st team, 2nd team, and 3rd team designations. This means that the model overall performs well.

0.2.6.2 Interpretation

Posterior estimates indicate that points scored and Win Share ratio are the strongest predictors of All-NBA selection. A one standard deviation increase in points corresponds to a large increase in the log-odds of selection ($\beta = 4.15$, 95% Cred. Int.: 1.82–7.97), while Win Share ratio also exhibits a substantial positive effect ($\beta = 2.86$, 95% Cred. Int: 0.85–5.94). In contrast, several defensive and playing time variables show posterior intervals overlapping zero, suggesting weaker or uncertain contributions once scoring and overall value metrics are included.

```
Running "C:/PROGRA~1/R/R-44~1.1/bin/x64/Rcmd.exe" SHLIB foo.c
using C compiler: 'gcc.exe (GCC) 13.3.0'
gcc  -I"C:/PROGRA~1/R/R-44~1.1/include" -DNDEBUG   -I"C:/Users/luxma/AppData/Local/R/win-li
cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
In file included from C:/Users/luxma/AppData/Local/R/win-library/4.4/RcppEigen/include/Eige
                 from C:/Users/luxma/AppData/Local/R/win-library/4.4/RcppEigen/include/Eige
                 from C:/Users/luxma/AppData/Local/R/win-library/4.4/StanHeaders/include/st
                 from <command-line>:
C:/Users/luxma/AppData/Local/R/win-library/4.4/RcppEigen/include/Eigen/src/Core/util/Macros
  679 | #include <cmath>
      |          ^~~~~~~
compilation terminated.
```

```
make: *** [C:/PROGRA~1/R/R-44~1.1/etc/x64/Makeconf:289: foo.o] Error 1


SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 1).
Chain 1:
Chain 1: Gradient evaluation took 8.4e-05 seconds
Chain 1: 1000 transitions using 10 leapfrog steps per transition would take 0.84 seconds.
Chain 1: Adjust your expectations accordingly!
Chain 1:
Chain 1:
Chain 1: Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 1: Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 1: Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 1: Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 1: Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 1: Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 1: Iteration: 1001 / 2000 [ 50%]  (Sampling)
Chain 1: Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 1: Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 1: Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 1: Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 1: Iteration: 2000 / 2000 [100%]  (Sampling)
Chain 1:
Chain 1:  Elapsed Time: 1.148 seconds (Warm-up)
Chain 1:                1.038 seconds (Sampling)
Chain 1:                2.186 seconds (Total)
Chain 1:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 2).
Chain 2:
Chain 2: Gradient evaluation took 2.6e-05 seconds
Chain 2: 1000 transitions using 10 leapfrog steps per transition would take 0.26 seconds.
Chain 2: Adjust your expectations accordingly!
Chain 2:
Chain 2:
Chain 2: Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 2: Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 2: Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 2: Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 2: Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 2: Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 2: Iteration: 1001 / 2000 [ 50%]  (Sampling)
Chain 2: Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 2: Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 2: Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 2: Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 2: Iteration: 2000 / 2000 [100%]  (Sampling)
```

```
Chain 2:
Chain 2:  Elapsed Time: 1.068 seconds (Warm-up)
Chain 2:                1.219 seconds (Sampling)
Chain 2:                2.287 seconds (Total)
Chain 2:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 3).
Chain 3:
Chain 3: Gradient evaluation took 2.6e-05 seconds
Chain 3: 1000 transitions using 10 leapfrog steps per transition would take 0.26 seconds.
Chain 3: Adjust your expectations accordingly!
Chain 3:
Chain 3:
Chain 3: Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 3: Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 3: Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 3: Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 3: Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 3: Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 3: Iteration: 1001 / 2000 [ 50%]  (Sampling)
Chain 3: Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 3: Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 3: Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 3: Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 3: Iteration: 2000 / 2000 [100%]  (Sampling)
Chain 3:
Chain 3:  Elapsed Time: 1.171 seconds (Warm-up)
Chain 3:                0.919 seconds (Sampling)
Chain 3:                2.09 seconds (Total)
Chain 3:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 4).
Chain 4:
Chain 4: Gradient evaluation took 3.1e-05 seconds
Chain 4: 1000 transitions using 10 leapfrog steps per transition would take 0.31 seconds.
Chain 4: Adjust your expectations accordingly!
Chain 4:
Chain 4:
Chain 4: Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 4: Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 4: Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 4: Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 4: Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 4: Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 4: Iteration: 1001 / 2000 [ 50%]  (Sampling)
Chain 4: Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 4: Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 4: Iteration: 1600 / 2000 [ 80%]  (Sampling)
```

```
Chain 4: Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 4: Iteration: 2000 / 2000 [100%]  (Sampling)
Chain 4:
Chain 4:  Elapsed Time: 1.063 seconds (Warm-up)
Chain 4:                1.131 seconds (Sampling)
Chain 4:                2.194 seconds (Total)
Chain 4:

Probability of Direction

Parameter   |     pd
--------------------
(Intercept) |   100%
ratioWS     | 99.83%
G           | 98.98%
PTS         |   100%
MP          | 74.52%
BLK         | 88.00%
STL         | 86.48%
PF          | 88.45%

 Family: bernoulli
  Links: mu = logit
Formula: is_all_nba ~ ratioWS + G + PTS + MP + BLK + STL + PF
   Data: player_df_scaled (Number of observations: 150)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept    -8.76      3.00   -16.11    -4.60 1.00     1937     1303
ratioWS       2.74      1.21     0.76     5.64 1.00     2427     2244
G            -1.93      0.96    -4.09    -0.27 1.00     2328     2129
PTS           4.11      1.53     1.82     7.86 1.00     1861      835
MP           -0.76      1.18    -3.45     1.29 1.00     3294     2848
BLK           0.75      0.70    -0.31     2.36 1.00     1060      406
STL           0.73      0.67    -0.38     2.16 1.00     2362     2765
PF           -0.59      0.54    -1.79     0.26 1.00     2114     1783

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).

# A tibble: 15 x 45
   predicted_rank prob_all_nba    Rk Player    Age Team  Pos       G    GS    MP
             <int>        <dbl> <dbl> <chr>   <dbl> <chr> <chr> <dbl> <dbl> <dbl>
 1              1        1.000     1 Shai G~    26 OKC   PG       76    76  2598
```

```
 2              2        0.999     3 Nikola~    29 DEN   C        70    70  2571
 3              3        0.998     4 Gianni~    30 MIL   PF       67    67  2289
 4              4        0.954     5 Jayson~    26 BOS   PF       72    72  2624
 5              5        0.891    18 Jalen ~    28 NYK   PG       65    65  2301
 6              6        0.888    14 Stephe~    36 GSW   PG       70    70  2252
 7              7        0.874     2 Anthon~    23 MIN   SG       79    79  2871
 8              8        0.825    16 LeBron~    40 LAL   SF       70    70  2444
 9              9        0.780    11 Karl-A~    29 NYK   C        72    72  2517
10             10        0.738    17 Donova~    28 CLE   SG       71    71  2232
11             11        0.718    24 Jalen ~    23 OKC   SG       69    69  2237
12             12        0.714    38 Tyrese~    24 IND   PG       73    73  2451
13             13        0.589    47 Evan M~    23 CLE   PF       71    71  2167
14             14        0.571     9 Cade C~    23 DET   PG       70    70  2452
15             15        0.492    10 James ~    35 LAC   PG       79    79  2789
# i 35 more variables: FG <dbl>, FGA <dbl>, `FG%` <dbl>, `3P` <dbl>,
#   `3PA` <dbl>, `3P%` <dbl>, `2P` <dbl>, `2PA` <dbl>, `2P%` <dbl>,
#   `eFG%` <dbl>, FT <dbl>, FTA <dbl>, `FT%` <dbl>, ORB <dbl>, DRB <dbl>,
#   TRB <dbl>, AST <dbl>, STL <dbl>, BLK <dbl>, TOV <dbl>, PF <dbl>, PTS <dbl>,
#   `Trp-Dbl` <dbl>, Awards <chr>, `Player-additional` <chr>, all_nba <chr>,
#   is_all_nba <dbl>, ratioWS <dbl>, ratioOWS <dbl>, ratioDWS <dbl>,
#   ratioWSPer48 <dbl>, ratioOBPM <dbl>, ratioDBPM <dbl>, ratioBPM <dbl>, ...
```

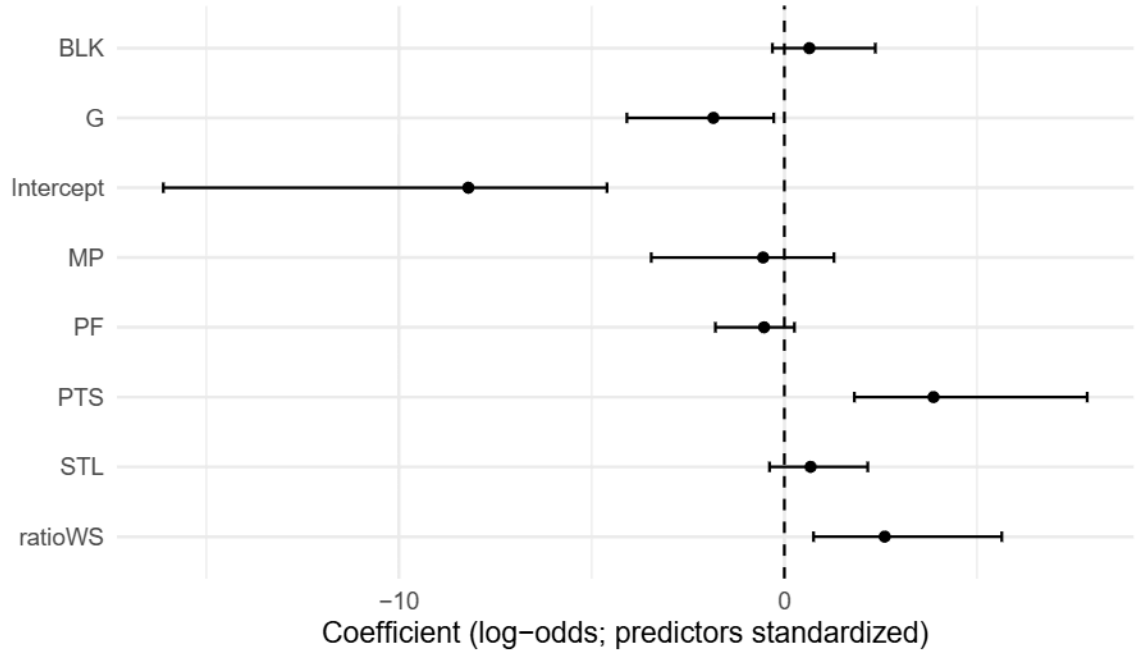


Figure 2: Posterior medians and 95% credible intervals for the refined horseshoe logistic regression.